

Predicting Cancer Types from Gene Expression Data Using Deep Learning

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This project aims to build a deep-learning model that predicts cancer types from gene-expression profiles using RNA-seq data from The Cancer Genome Atlas (TCGA). Gene-expression signatures capture the activity of thousands of genes simultaneously and different cancers exhibit distinct molecular patterns that can be learned by machine-learning models. Accurate cancer classification from expression data is clinically relevant because it can support the diagnosis, reveal the underlying biology and potentially guide treatment decisions.

The idea of using neural networks for cancer classification is well-established in the literature. A landmark study by Khan et al. (2001) showed that artificial neural networks can accurately distinguish four childhood cancers (SRBCTs) using microarray gene-expression profiles. Their model achieved perfect accuracy on both training and blinded test samples, demonstrating that gene-expression data contains strong diagnostic signals and that neural networks are effective tools for capturing these patterns. More recently, Kim, Yu & Lee (2020) applied a feed-forward neural network to single-cell RNA-seq data and found that it outperformed classical machine-learning methods such as SVMs and Random Forests. Their results show that deep learning is particularly well-suited for high-dimensional biological data, even when the data is noisy or sparse. Together, these studies provide strong support for our approach: FNNs work well on gene expression and cancer classification from expression is a valid and widely studied task.

In our project, each TCGA sample will be represented as a high-dimensional vector of gene-expression values. We will begin with exploratory data analysis, including normalization, variance filtering, and dimensionality reduction (PCA) to understand the structure of the dataset. We will then design a feed-forward neural network (FNN) in PyTorch, using several dense layers with ReLU activation, dropout and weight decay to prevent overfitting. The model will be trained to classify samples into cancer types and we will experiment with different architectures, hidden-layer sizes and regularization strategies. Performance will be evaluated using accuracy, confusion matrices and training/validation curves.

Finally, we will discuss challenges such as high dimensionality, class imbalance and overfitting. Overall, this project demonstrates how deep learning can be applied to real biomedical data and how the model design choices influence the performance in a high-dimensional cancer-classification task.

Bong-Hyun Kim, Kijin Yu, Peter C W Lee, Cancer classification of single-cell gene expression data by neural network, *Bioinformatics*, Volume 36, Issue 5, March 2020, Pages 1360–1366, <https://doi.org/10.1093/bioinformatics/btz772>

Khan, J., Wei, J., Ringnér, M. et al. Classification and diagnostic prediction of cancers using gene expression profiling and artificial neural networks. *Nat Med* 7, 673–679 (2001). <https://doi.org/10.1038/89044>